Mat's Science Explainer Pack · Vol. 2

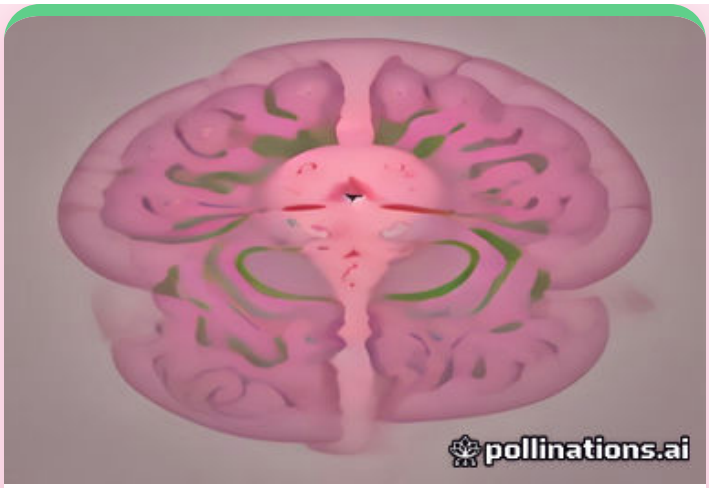
16 bite-size explainers · Neuroscience · Animals · Astrophysics · Science



🧠 Neuroscience · #1

Dopamine: The Prediction Molecule

Dopamine is often called the pleasure chemical, but that nickname misses its real job. Dopamine neurons, clustered in small midbrain areas such as the ventral tegmental area, mostly signal surprise about rewards. In classic experiments in the 1990s, Wolfram Schultz recorded these neurons in monkeys. When an unexpected drop of juice arrived, dopamine neurons burst with activity. After the monkey learned that a light always came before the juice, the burst moved to the light, and the juice itself caused little response. If the light appeared but the juice never came, the neurons dipped below their normal firing rate. This pattern matches a reward prediction error: actual reward minus expected reward. Computer scientists had already used the same math to train learning machines, so the discovery linked brains and artificial intelligence. Prediction errors teach the brain what is worth pursuing and drive motivation, the wanting rather than the liking. That helps explain why unpredictable rewards, like slot machines or social media notifications, are so compelling. Dopamine also controls movement. In Parkinson's



🧠 Neuroscience · #2

Phantom Limbs and the Body Map

Many people who lose an arm or leg still feel it afterward, sometimes for years. They may sense it moving, itching, or painfully cramped. This phantom limb is not imagination; it comes from the brain's maps of the body. Along a strip of cortex called the somatosensory cortex, each body part has its own patch, arranged in order like a distorted figure called the homunculus, with huge hands and lips because those areas are so sensitive. When a limb is lost, its patch stops receiving input, but it does not stay silent. Neighboring regions can invade it. In the hand map, the face area sits right next door. Neuroscientist V.S. Ramachandran found that touching some amputees on the cheek made them feel touch on their missing fingers. Phantom pain can be severe, and Ramachandran invented a simple treatment: the mirror box. A patient places the healthy arm beside a mirror so its reflection appears where the missing arm would be. Moving the real arm makes the phantom seem to move, which can relieve cramping in some people. Phantom limbs show that our sense of having a body is actively built by the brain.

disease, dopamine neurons in the substantia nigra die, causing tremor and stiffness, which is why the drug L-dopa helps.



🧠 Neuroscience · #3

Why You Can't Tickle Yourself

Try tickling the sole of your own foot and you will feel almost nothing, yet the same touch from a friend can make you squirm. The reason is that your brain predicts the results of your own movements. Whenever you move, the motor system sends a copy of the command, called an efference copy, to other brain areas. The cerebellum, at the back of the brain, uses that copy to forecast the sensations the movement will cause, then subtracts the forecast from what actually arrives. Expected sensations are turned down, so self-touch feels weaker and less important. Researchers at University College London, led by Sarah-Jayne Blakemore, tested this with a robot. When people moved a lever that made a robot stroke their palm, it felt less ticklish. When the scientists added a delay of a fraction of a second or changed the direction, the touch felt more ticklish, because the prediction no longer matched. This canceling system helps us notice what matters: a spider on your arm stands out, while your own sleeves brushing your skin fade away. Interestingly, some people with schizophrenia can tickle themselves, suggesting their self-prediction works differently.



🧠 Neuroscience · #4

The Gut-Brain Connection

Your gut contains around 500 million neurons, more than your spinal cord, forming the enteric nervous system, sometimes called the second brain. It can run digestion on its own, squeezing food along and controlling enzyme release, but it also talks constantly with the brain. The main phone line is the vagus nerve, and about 80 percent of its fibers carry signals upward, from gut to brain, rather than downward. The gut is also a chemical factory. Roughly 90 percent of the body's serotonin is made there, though this gut serotonin mostly controls intestinal movement and does not cross into the brain. The trillions of microbes living in your intestines join the conversation too. They break down fiber into short-chain fatty acids, influence immune cells, and produce molecules that can affect nerves. In experiments, mice raised without any gut microbes showed exaggerated stress responses, and transferring microbes between mice could shift some behaviors. Research in humans is earlier and harder to interpret, and claims about probiotics curing anxiety go beyond the evidence. Still, the familiar feelings of butterflies in your

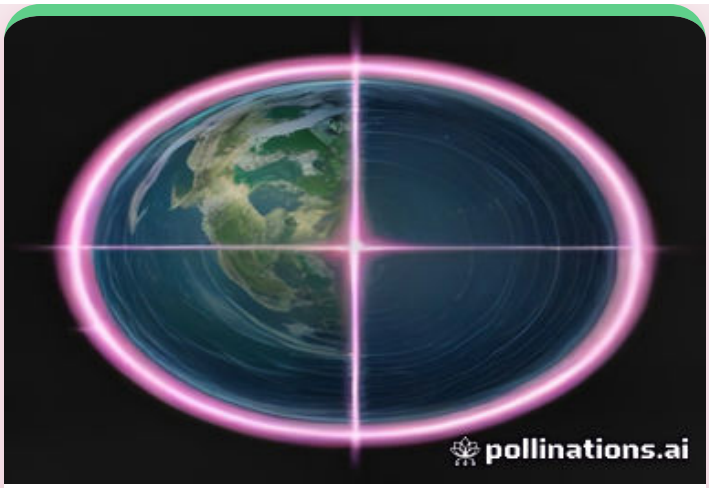
stomach or a gut instinct have a real biological basis in this two-way link.



🐙 Animal Kingdom · #5

The Mantis Shrimp's Super Punch

The mantis shrimp is a colorful crustacean, not a true shrimp, that lives in burrows on tropical reefs. Smashers, one type of mantis shrimp, carry a pair of club-shaped limbs that strike prey with one of the fastest movements in the animal kingdom. The peacock mantis shrimp's club accelerates more than 10,000 times the acceleration of gravity and reaches about 23 meters per second, fast enough to crack snail shells and even aquarium glass. The trick is a spring. A saddle-shaped structure stores energy like a bow while a latch holds the club back, then releases it all at once. The strike is so fast that it boils the water in its path, creating collapsing bubbles called cavitation, so prey is hit twice, once by the club and once by the bubble. The club itself is a marvel of materials science, with layers of mineral arranged in a spiral pattern that stops cracks spreading, inspiring new tough materials. Mantis shrimp eyes are just as strange, with 12 to 16 types of photoreceptor compared with our three. Yet tests suggest they are worse than us at telling similar colors apart, perhaps



🐙 Animal Kingdom · #6

How Birds Navigate by Magnetism

Every year billions of birds migrate thousands of kilometers, and many find their way across oceans at night. They use several tools: the position of the sun, star patterns, landmarks, smells, and Earth's magnetic field. The magnetic sense is the most mysterious. In the 1960s, German scientists showed that caged European robins, restless to migrate, hopped in the correct direction even with no view of the sky, and changed direction when the magnetic field around them was rotated. Birds seem to use a magnetic inclination compass. Rather than sensing north and south like a needle, they detect the angle at which field lines dip toward the ground, which tells them whether they are heading toward the pole or the equator. A leading idea is that this sense works through quantum chemistry in the eye. Proteins called cryptochromes, especially cryptochrome 4, form pairs of molecules with linked electrons when struck by light, and Earth's weak field may influence how those pairs react. If so, birds might literally see the magnetic field as a pattern over their vision. Remarkably, robins' compass

recognizing colors quickly instead of comparing them.

can be disrupted by weak radio-frequency noise from electronics, which supports the quantum explanation.



🐙 Animal Kingdom · #7

Naked Mole-Rats: Rule Breakers

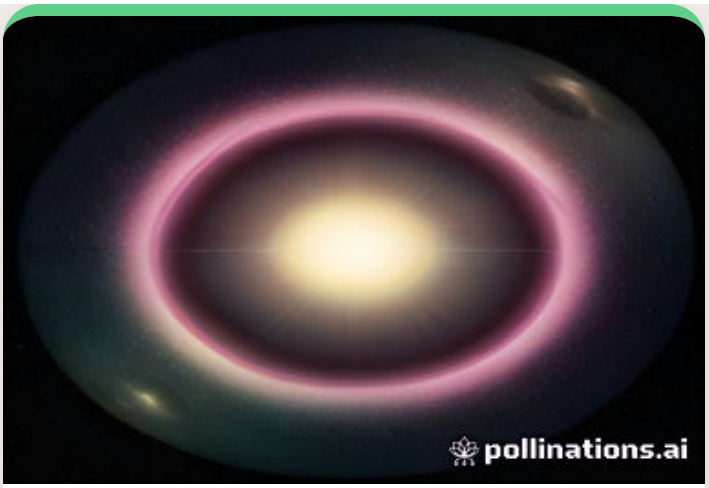
Naked mole-rats look like wrinkly pink sausages with teeth, but biologists consider them among the most surprising mammals alive. They live underground in East Africa in colonies of up to around 300, organized like an ant or bee nest. A single queen breeds with one to three males, while the rest work as diggers, food gatherers, and soldiers. This system, called eusociality, is extremely rare in mammals. Their bodies break rules too. A mouse-sized rodent normally lives about four years, but naked mole-rats can live more than 37 years, and their chance of dying does not seem to rise with age the way it does in most mammals. Cancer is extremely rare in them, possibly helped by a very long form of a sugary molecule called hyaluronan that keeps cells from crowding. They can survive 18 minutes with no oxygen by switching to burning fructose, a trick more like a plant's. Their skin lacks certain pain signals, so acid does not sting them. They are nearly cold-blooded, huddling together to stay warm. Scientists study them hoping to learn secrets about aging and cancer that might someday help people.



🐙 Animal Kingdom · #8

The Immortal Jellyfish

Turritopsis dohrnii, a jellyfish smaller than a fingernail, has earned the nickname the immortal jellyfish. Like most jellyfish, it begins as a larva that settles on the seafloor and grows into a polyp, a small stalk that buds off free-swimming adults called medusae. Normally the story ends there: medusae reproduce and die. But when Turritopsis is starved, injured, or stressed, it can do something astonishing. The adult sinks, shrinks into a blob, and transforms back into a polyp, which then produces new medusae. It is as if a butterfly could turn back into a caterpillar. The key process is transdifferentiation, in which specialized cells change directly into different cell types. In 2022, a team led by Spanish scientists compared its genome with a close relative and found extra copies of genes for DNA repair and protecting the ends of chromosomes, called telomeres. In theory, this cycle could repeat indefinitely, so the species is biologically immortal. In reality, individuals are eaten by predators or killed by disease all the time, so no single jellyfish lives forever. Kyoto University researcher Shin Kubota famously kept them in his lab and watched them rejuvenate many times.



🌌 Astrophysics · #9

Finding Planets Around Other Stars

In 1995, Michel Mayor and Didier Queloz discovered the first planet orbiting a Sun-like star, 51 Pegasi b, earning a Nobel Prize. Today more than 5,000 exoplanets are confirmed. Planets are tiny and faint next to their stars, so astronomers mostly find them indirectly. The transit method watches for a star dimming slightly when a planet passes in front of it. A Jupiter-sized planet blocks about 1 percent of a Sun-like star's light, while an Earth-sized one blocks just 0.01 percent. NASA's Kepler telescope used this method to find thousands of planets, and its successor TESS scans nearly the whole sky. The radial velocity method detects the star wobbling as the planet's gravity tugs it, which shifts its light slightly toward blue and then red. Combining both methods gives a planet's size and mass, and therefore its density, revealing whether it is rocky or gassy. The results surprised everyone. Many systems contain hot Jupiters orbiting closer than Mercury, and the most common planets are super-Earths and mini-Neptunes, sizes missing from our own solar system. The James Webb Space Telescope can now study starlight filtering through



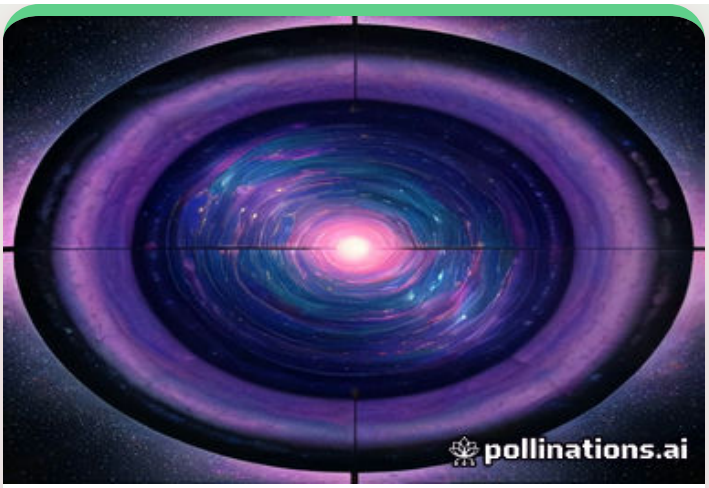
🌌 Astrophysics · #10

Dark Matter: The Invisible Majority

Everything we can see, from stars to gas to people, makes up only about 15 percent of the matter in the universe. The rest is dark matter, which does not emit, absorb, or reflect light, and reveals itself only through gravity. The first hint came in 1933, when Fritz Zwicky noticed that galaxies in the Coma Cluster moved too fast to be held together by their visible mass. In the 1970s, Vera Rubin and Kent Ford measured how fast stars orbit in spiral galaxies. Stars far from the center should move slower, as outer planets do around the Sun, but instead they moved about as fast as inner ones. Galaxies seemed wrapped in huge invisible halos. More evidence followed. Massive clusters bend light from background galaxies, an effect called gravitational lensing, revealing far more mass than we see. In the Bullet Cluster, two colliding clusters left their hot gas in the middle while most of the mass, mapped by lensing, passed straight through. Patterns in the cosmic microwave background agree too. Yet decades of experiments deep underground have not detected a dark

some planets' atmospheres, detecting molecules such as water and carbon dioxide.

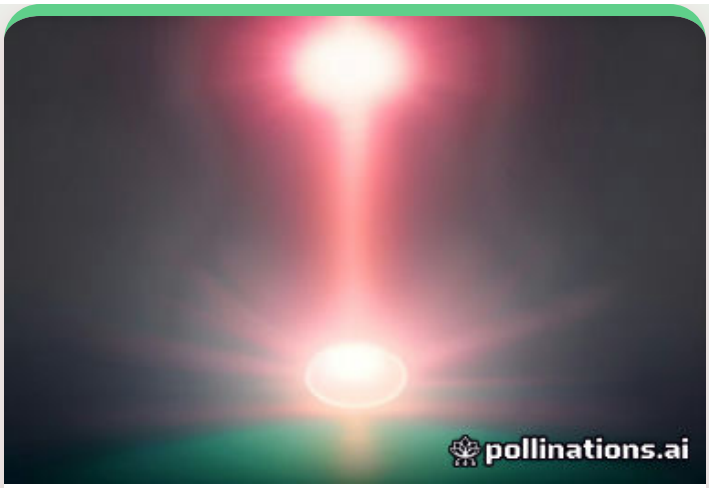
matter particle directly, so what it actually is remains one of physics' biggest mysteries.



🌌 Astrophysics · #11

The Oldest Light in the Universe

Point a sensitive radio telescope at any patch of sky and you will detect a faint glow coming from every direction. This is the cosmic microwave background, the oldest light we can see. For about the first 380,000 years after the Big Bang, the universe was a hot, dense fog of charged particles. Light could not travel far before bouncing off free electrons. As space expanded and cooled to about 3,000 °C, electrons joined nuclei to form neutral atoms, and the fog suddenly cleared. The light released at that moment has been traveling ever since. Over 13.8 billion years, the expansion of space has stretched its wavelengths a thousandfold, so it now appears as microwaves at just 2.7 degrees above absolute zero. Arno Penzias and Robert Wilson discovered it by accident in 1965 while trying to remove a mysterious hiss from their antenna, and won a Nobel Prize. Satellites such as COBE, WMAP, and Planck mapped tiny temperature differences of about one part in 100,000. These ripples were the seeds of galaxies, and their pattern lets cosmologists measure the universe's age, shape, and ingredients. Some old

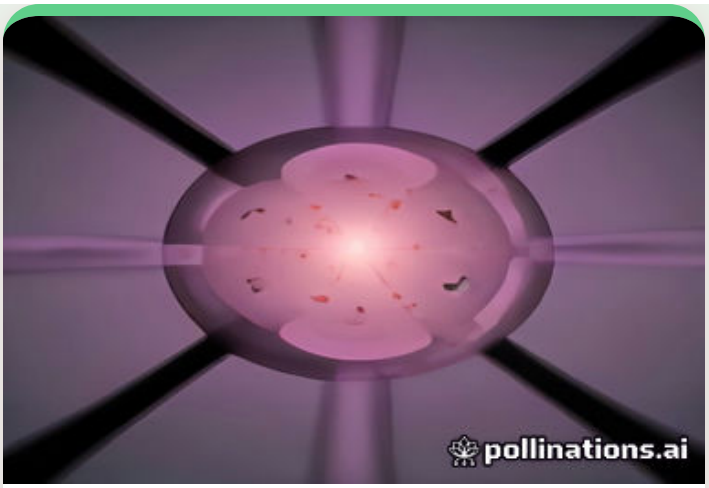


🌌 Astrophysics · #12

Sunlight's Long Journey

The sunlight warming your face left the Sun's surface only about eight minutes ago, but the energy it carries began its journey far earlier. Deep in the Sun's core, at about 15 million °C, hydrogen nuclei fuse into helium. Each second the Sun converts about 4 million tonnes of mass into energy, following Einstein's E equals m c squared. This energy starts as high-energy gamma rays, but it cannot simply fly out. The inner Sun is so dense that a photon travels only a tiny distance before being absorbed and re-emitted in a random direction. This drunkard's walk through the radiative zone is painfully slow. Estimates vary widely, but energy typically takes tens of thousands to over a hundred thousand years to cross it. Then, in the outer third of the Sun, energy is carried much faster by convection, as huge bubbles of hot gas rise, cool, and sink, like boiling soup. These bubbles create the grainy pattern called granulation on the surface. By the time the energy escapes from the photosphere, it has softened from gamma rays into mostly visible light. Interestingly, neutrinos made in the same fusion reactions zip straight out in about two seconds.

analog TV static was partly caused by this ancient light.



🔬 Science · #13

How Vaccines Train Your Immune System

Your immune system has two layers. The innate system attacks invaders quickly but generally, while the adaptive system learns to recognize specific germs and remembers them. A vaccine is a safe rehearsal for the adaptive system. It shows the body a harmless piece or version of a germ, called an antigen, without causing the disease. Different vaccines do this in different ways: weakened live viruses, inactivated germs, purified proteins, or genetic instructions such as mRNA, which tell your own cells to briefly make a viral protein before the instructions break down. Immune cells called B cells that happen to recognize the antigen multiply and produce antibodies, Y-shaped proteins that latch onto the germ and flag it for destruction. T cells help coordinate the response and kill infected cells. Crucially, some of these cells become long-lived memory cells. If the real germ ever arrives, they respond faster and stronger, often stopping illness before it starts. Vaccines also protect people who cannot be vaccinated, because when enough of a population is immune, a germ struggles to spread, a concept called herd



🔬 Science · #14

What Makes Lightning

A lightning bolt is a giant electric spark, and it begins inside towering thunderclouds called cumulonimbus. Strong updrafts carry water droplets high into freezing air, where they form ice crystals and soft hail pellets called graupel. As these particles collide in the turbulent cloud, electric charge is transferred between them. Small, light ice crystals tend to become positively charged and are carried to the top, while heavier graupel becomes negatively charged and gathers in the middle and lower cloud. Eventually the voltage difference grows to hundreds of millions of volts. Air is normally a good insulator, but when the electric field becomes strong enough, a faint channel of ionized air called a stepped leader zigzags downward in jumps of about 50 meters. As it nears the ground, an upward streamer rises from a tall object to meet it. When they connect, a powerful return stroke races upward, heating the channel to about 30,000 °C, around five times hotter than the Sun's surface. That superheated air expands explosively, creating thunder. Because light travels much faster than sound, you can estimate the distance: every three seconds between flash and thunder means roughly

immunity. Smallpox, which killed hundreds of millions, was declared eradicated in 1980 thanks to vaccination.

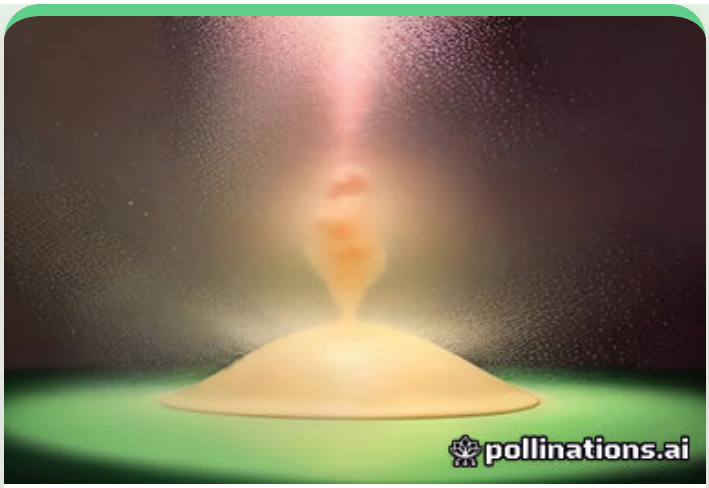
one kilometer. Most lightning actually stays within clouds.



🔬 Science · #15

CRISPR: Editing the Code of Life

CRISPR is a gene-editing tool borrowed from bacteria. Bacteria are constantly attacked by viruses, and many defend themselves by saving snippets of viral DNA in their own genome, in repeating sections called clustered regularly interspaced short palindromic repeats, or CRISPR. If the same virus returns, the bacterium makes RNA copies of those snippets. Each RNA guides a cutting protein, such as Cas9, to the matching viral DNA, which it slices apart. In 2012, Jennifer Doudna and Emmanuelle Charpentier showed that this system could be reprogrammed. By writing a custom guide RNA, scientists can direct Cas9 to almost any DNA sequence and cut it there. The cell then repairs the break, which can disable a gene or, if a template is supplied, insert a correction. The pair won the 2020 Nobel Prize in Chemistry. CRISPR is now used everywhere in biology research, and in 2023 the first CRISPR medicine, Casgevy, was approved to treat sickle cell disease and beta thalassemia by editing patients' own blood stem cells. Newer versions, called base editors and prime editors, can change single DNA letters without cutting both



🔬 Science · #16

Why Bread Rises

A loaf of bread is a small triumph of biology and chemistry. Baker's yeast, Saccharomyces cerevisiae, is a single-celled fungus. When mixed into dough, it feeds on sugars, some released from starch by enzymes in the flour. Without much oxygen, yeast ferments these sugars into carbon dioxide gas and ethanol. The gas needs something to trap it, and that is gluten. When flour is mixed with water, two proteins, glutenin and gliadin, link into a stretchy, elastic network. Kneading lines up and strengthens this network, so it can stretch like a balloon around millions of tiny gas bubbles. That is why wheat bread rises so well, while flours with little gluten, such as rice flour, make denser loaves unless other ingredients help. In the oven, the gas expands with heat, giving a final burst of rising called oven spring, before the yeast dies at around 60 °C. The alcohol evaporates, starches set, and proteins firm up, locking the airy structure in place. Meanwhile, the Maillard reaction between sugars and proteins, along with caramelization, browns the crust and creates hundreds of flavor compounds.

strands. The power also raises ethical questions, especially about editing human embryos.

Sourdough adds lactic acid bacteria, which provide a tangy taste.